CLASS 10 · JEE ADVANCED · INSTITUTE TOP 10 VS OUR TOP 10

Pragya G10 (EXI) Achievers

Part Test-1 · Paper 2 · where our toppers fall behind the institute's toppers

Exam date 09 Oct 2026
Campus HYD/Nallakunta DD Colony
Compared institute top 10 vs our top 10

On average the institute's top 10 scored ≈ 132.0 and our top 10 scored ≈ 129.7 – a gap of ≈ 2.3 marks per student: 1.8 in Maths and 0.9 in Physics.

Maths 67% of the gap | Physics 33% of the gap

7 of the institute's top 10 are our students (program rank 1, 2, 4, 5, 6, 7, 8), so this compares the institute's other 3 toppers with our next 3. Scores are rebuilt from each question's right / wrong / skipped counts with the paper's marking; approximate (≈) because multiple-correct questions give partial marks – our official average is 128.9.

≈ 132 / 176
institute top 10 – average score

≈ 130 / 176
our top 10 – average score

5 questions
where 2+ more of the institute top 10 got it right

4 questions
where 2+ more of ours got it right

1 Subject by subject

Average marks per student, and how the answers split into right, wrong and skipped. ■ Institute top 10 ■ Our top 10

SUBJECT	AVERAGE SCORE		GAP	RIGHT	WRONG	SKIPPED
Maths out of 60	≈ 43.7 ≈ 41.9		1.8 per student	77% vs 75%	17% vs 18%	6% vs 6%
Physics out of 56 = 60 minus Q25 (4)	≈ 38.3 ≈ 37.4		0.9 per student	67% vs 66%	22% vs 26%	11% vs 8%
Chemistry out of 60	≈ 50.0 ≈ 50.4		−0.4 per student	85% vs 85%	14% vs 13%	2% vs 2%

"Out of" = the subject's marks on this paper minus the questions left out by the final key (marks of each left-out question in brackets). Percentages are of all answers in that subject (questions × students), institute first, ours second.

Key takeaways

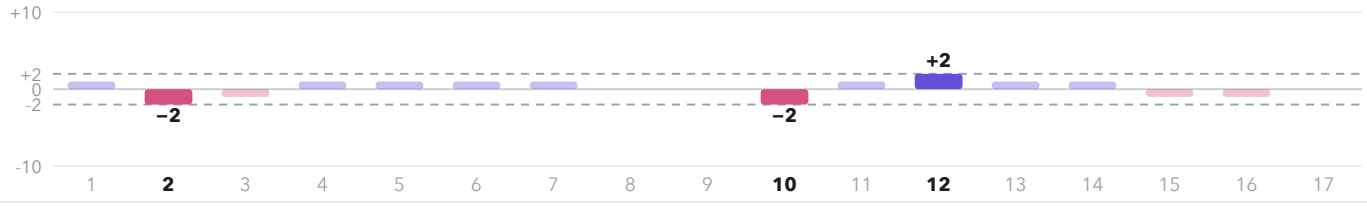
1. **7 of the institute's top 10** are our own students (program rank 1, 2, 4, 5, 6, 7, 8), so the two groups mostly overlap: at most 3 answers can differ on any question, so the gap line is set at 2 instead of 4.
2. **Maths** is where we fall furthest behind: ≈ **1.8** marks per student (67% of the total gap).
3. **5 questions** were answered correctly by at least 2 more of the institute's top 10 than ours. Biggest: Q46 (5 vs 2), Q12 (8 vs 6), Q27 (8 vs 6).
4. Where we did better: Q2 (5 vs 3), Q10 (10 vs 8), Q21 (6 vs 4), Q51 (5 vs 3).

2 Question-by-question gap

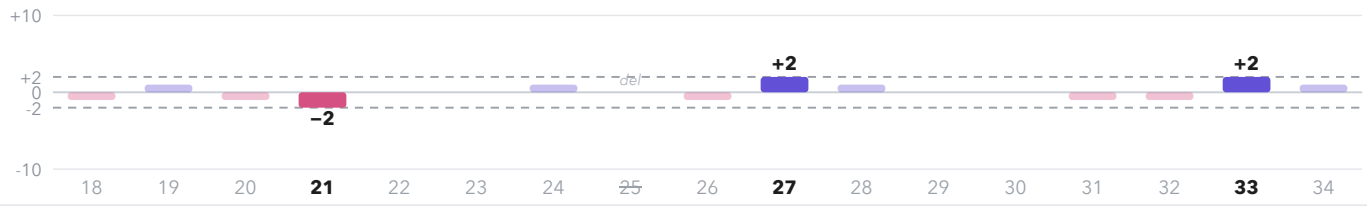
For each question: how many more of the institute's top 10 got it right than ours. Bars above the line = the institute did better; below = we did better. Gaps of 2+ (dashed line) are labelled.

Institute top 10 got more right Our top 10 got more right ---- Gap of 2

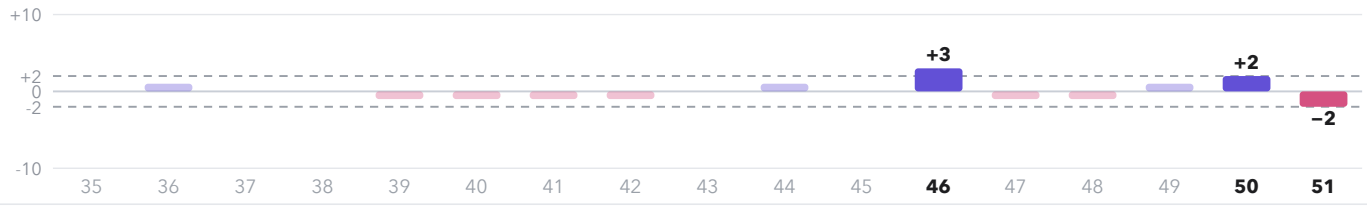
Maths — **1** question with a gap of 2+ · gap ≈ **1.8** marks per student



Physics — **2** questions with a gap of 2+ · gap ≈ **0.9** marks per student



Chemistry — **2** questions with a gap of 2+ · gap ≈ **-0.4** marks per student



3 Chapter gaps

Average marks per student on each chapter's questions, institute top 10 vs ours. Subjects are ordered from the biggest gap to the smallest; within each subject, largest chapter gap first.

CHAPTER	QS	INSTITUTE VS OURS	GAP PER STUDENT	GAP QUESTIONS
1. Maths gap ≈ **1.8** marks per student · **1** question with a gap of 2+				
Differential Equations	5	11.2 vs 9.0	≈ 2.2	Q12
Indefinite Integration	4	13.4 vs 12.7	≈ 0.7	–
Area Under Curves	3	6.7 vs 6.9	−≈ 0.2	–
Definite Integration	5	12.4 vs 13.3	−≈ 0.9	–
2. Physics gap ≈ **0.9** marks per student · **2** questions with a gap of 2+				
Electrostatics	6	15.5 vs 13.8	≈ 1.7	Q33
Current Electricity	2	6.4 vs 6.6	−≈ 0.2	–
Kinematics	8	16.4 vs 17.0	−≈ 0.6	Q27
3. Chemistry gap ≈ **-0.4** marks per student · **2** questions with a gap of 2+				
Aldehydes & Ketones	6	18.2 vs 18.2	≈ 0.0	–
Carboxylic Acids	4	10.0 vs 10.0	≈ 0.0	Q50
Alcohols, Phenols & Ethers	7	21.8 vs 22.2	−≈ 0.4	Q46

4 Biggest gaps – question list

Questions where at least 2 more of the institute's top 10 got the right answer than ours, one table per subject. The cropped question for each is in section 5. With 7 of the 10 students shared, at most 3 answers can differ on any question, so the line is set at 2 instead of 4.

Maths

1 question with a gap of 2+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q12	**Differential Equations** – Solve a first-order DE and match its constants	Integer · +4 / 0	8/10	6/10 · 4 · 0	+2

Physics

2 questions with a gap of 2+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q27	**Kinematics** – Projection angle that maximises area under the trajectory	Integer · +4 / 0	8/10	6/10 · 4 · 0	+2
Q33	**Electrostatics** – Flux through a ring – solid angle	Paragraph · +3 / 0	7/10	5/10 · 5 · 0	+2

Chemistry

2 questions with a gap of 2+

Q	CHAPTER – WHAT IT TESTS	TYPE	INSTITUTE RIGHT	OURS RIGHT · WRONG · SKIPPED	GAP
Q46	**Alcohols, Phenols & Ethers** – Identifying polar aprotic solvents	Integer · +4 / 0	5/10	2/10 · 5 · 3	+3
Q50	**Carboxylic Acids** – Molar mass of an acid from NaOH titration	Paragraph · +3 / 0	9/10	7/10 · 3 · 0	+2

Where our top 10 did better

Q2 5 vs 3 right · **Q10** 10 vs 8 right · **Q21** 6 vs 4 right · **Q51** 5 vs 3 right

How to read this report. "Institute top 10" = the *Top 10 Students* rows of the institute's question-wise R/W/U sheet for this test (program ranks 1–10 in Pragya G10 (EXI) Achievers, all campuses). "Our top 10" = this campus's top 10 by program rank. A gap of +5 on a question means 5 more of the institute's top 10 got it right than ours.

5 Maths – biggest gaps

1 question, largest gap first
cropped from the question paper

Q12

Differential Equations – **Solve a first-order DE and match its constants**

Institute top 10: **8 right**, 2 wrong, 0 skipped · Our top 10: **6 right**, 4 wrong, 0 skipped

Maths · Integer · marking +4 / 0

Gap +2

12. The differential equation $\left(xy^3\left(1+\cos x\right)-y\right)dx+xdy=0$ represents the curve $\frac{x^2}{ay^2}=\frac{x^3}{b}+x^2\sin x+cx\cos x-d\sin x+k$. Then a+b+c+d is equal to (where 'k' is arbitrary constant).

Ours wrong: Dharshan Raj S, Thonta Harshavardhan, Gajavali Anjaneyulu, Yapala Dhanush Reddy

Ours skipped: –

Physics – biggest gaps

2 questions, largest gap first
cropped from the question paper

Q27

Kinematics – Projection angle that maximises area under the trajectory

Institute top 10: **8 right**, 2 wrong, 0 skipped · Our top 10: **6 right**, 4 wrong, 0 skipped

Physics · Integer · marking +4 / 0

Gap +2

27. A projectile is thrown with a speed 10 m/s, at an angle θ with the horizontal. For a certain value of angle of projection $\theta = \frac{(n+1)\pi}{12}$ the area bound by the trajectory with horizontal is maximum value of n is (g=10 m/s^2). Assume no airdrag.

Ours wrong: Dharshan Raj S, Mano Nischal Reddy D, Vihaan Bussari, Gajavali Anjaneyulu

Ours skipped: –

Q33

Electrostatics – Flux through a ring – solid angle

Institute top 10: **7 right**, 3 wrong, 0 skipped · Our top 10: **5 right**, 5 wrong, 0 skipped

Physics · Paragraph · marking +3 / 0

Gap +2

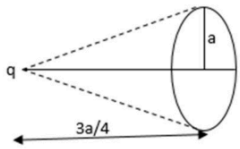
PARAGRAPH-II

The electric flux linked with a surface is the number of electric field lines associated with that area. Gauss law helps determine the flux

$$\oint \vec{E}.\vec{ds} = \frac{Q_{inc}}{\varepsilon_0}$$

For an open surface the flux will change depending on the solid angle. Considering this answer the following questions.

33.



The flux associated with the ring due to the charge q is given by Xq/ε$_0$. The value of X is approx.

Ours wrong: Poli Karthikeya Reddy, Shiva Sai Jyothiraditya Kokkonda, Vihaan Bussari, Gajavali Anjaneyulu, Yapala Dhanush Reddy

Ours skipped: –

Chemistry – biggest gaps

2 questions, largest gap first
cropped from the question paper

Q46

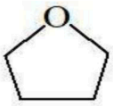
Alcohols, Phenols & Ethers – **Identifying polar aprotic solvents**

Institute top 10: **5 right**, 4 wrong, 1 skipped · Our top 10: **2 right**, 5 wrong, 3 skipped

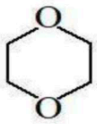
Chemistry · Integer · marking +4 / 0

Gap +3

46. How many of the following are polar aprotic solvents?



Diethyl ether, Benzene, Ethyl alcohol,



Petrol

[P] [Q] [R] [S] [T] [U]

Acetone DMSO (dimethyl sulphoxide DMF (dimethyl formamide formic acid

[Q] [W] [X] [Y]

Ours wrong: Dharshan Raj S, Thonta Harshavardhan, K Ranveer Reddy, Mano Nischal Reddy D, Vihaan Bussari

Ours skipped: Shiva Sai Jyothiraditya Kokkonda, Gajavali Anjaneyulu, Yapala Dhanush Reddy

Q50

Carboxylic Acids – **Molar mass of an acid from NaOH titration**

Institute top 10: **9 right**, 1 wrong, 0 skipped · Our top 10: **7 right**, 3 wrong, 0 skipped

Chemistry · Paragraph · marking +3 / 0

Gap +2

PARAGRAPH-II

Carboxylic acids are organic compounds containing the carbonyl(>C=O) and hydroxyl (-OH) functional groups combined as a carboxyl group(-COOH). Due to resonance stabilization of the carboxylate anion, carboxylic acids exhibit higher acidity compared to alcohols and phenols. They participate in several characteristic reactions, including nucleophilic acyl substitution (esterification, amide formation, acid halide synthesis), decarboxylation, and reduction to primary alcohols. Furthermore, the presence of the –COOH group strongly influences physical properties, enabling extensive intermolecular hydrogen bonding that results in higher boiling points and stable dimeric structures in non-polar solvents.

50. A 0.600 g sample of a pure monoprotic carboxylic acid (R-COOH) requires 25.00 mL of 0.200 M NaOH for complete neutralization. Calculate the molar mass of the carboxylic acid?

Ours wrong: Dharshan Raj S, Vihaan Bussari, Gajavali Anjaneyulu

Ours skipped: –